# Trigonometry

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## Cosine and sine

**Definition 0.1** (Cosine and sine)**.** The *unit circle* is the set of points $(x_1, x_2)$ in the plane with $x_1^2 + x_2^2 = 1$, and $\pi \approx 3.14159$ is half of its circumference. For a real number $\theta$, start at the point $(1, 0)$ and move a distance $|\theta|$ along the unit circle, counterclockwise if $\theta \geq 0$ and clockwise if $\theta < 0$. The point reached is $(\cos\theta, \sin\theta)$: its first coordinate is the **cosine** of $\theta$, and its second coordinate is the **sine** of $\theta$.

**Example 0.1** (Cosines and sines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\cos 0 = 1$ and $\sin 0 = 0$.
- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\cos\pi = -1$ and $\sin\pi = 0$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$.
- $\theta = \pi/4$ goes an eighth of the way round, to the point of the circle with $x_1 = x_2 > 0$. There $x_1^2 + x_1^2 = 1$, so $x_1^2 = \frac{1}{2}$ and $x_1 = \frac{1}{\sqrt{2}} \approx 0.7071$:

$$\begin{aligned} \cos(\pi/4) &= \sin(\pi/4) \\ &= \tfrac{1}{\sqrt{2}}. \end{aligned}$$

Angles are also measured in *degrees*, with a full turn of $2\pi$ equal to 360 degrees, so $\pi/2$ is 90 degrees and $\pi/4$ is 45 degrees.

**Theorem 0.1** (The Pythagorean identity)**.** *For every real number $\theta$,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Proof

*Proof.* The point $(\cos\theta, \sin\theta)$ lies on the unit circle (Definition 0.1), so its coordinates satisfy $x_1^2 + x_2^2 = 1$. □

**Example 0.2** (The unit circle and the angle $\pi/3$)**.** Take $\theta = \pi/3$, one sixth of the way round the circle. Figure 1 shows the point reached, $(\cos\theta, \sin\theta)$, and its two coordinates. The point is a vertex of a regular hexagon inscribed in the unit circle, whose sides have the same length as its radius, so the point is at distance 1 from $(1, 0)$. Writing $(x_1, x_2) = (\cos\theta, \sin\theta)$,

$$\begin{aligned} 1 &= (x_1 - 1)^2 + x_2^2 \\ &= x_1^2 + x_2^2 - 2x_1 + 1 \\ &= 2 - 2x_1, \end{aligned}$$

so $\cos(\pi/3) = x_1 = \frac{1}{2}$, and then $\sin(\pi/3) = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$. To three decimal places these are 0.5 and 0.866. As Theorem 0.1 says, the squared coordinates add to 1.

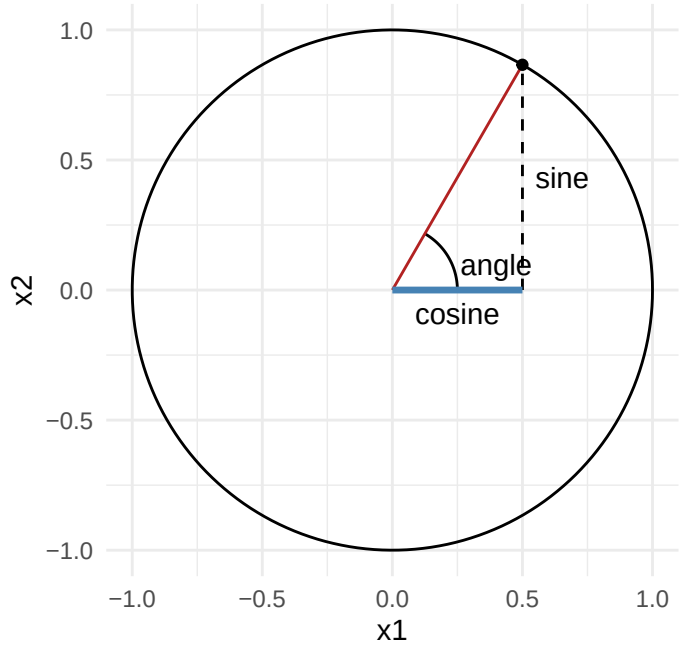


Figure 1: The unit circle with the point at angle $\pi/3$. Its horizontal coordinate is the cosine and its vertical coordinate is the sine.

**Theorem 0.2** (Cosine takes each value from $-1$ to 1 exactly once between 0 and $\pi$)**.** *The function* $\cos : [0, \pi] \to [-1, 1]$ *is decreasing, with* $\cos 0 = 1$ *and* $\cos \pi = -1$*, and takes each value in* $[-1, 1]$ *exactly once. On* $[0, \pi]$*,* $\sin \theta \geq 0$*.*

Proof

*Proof.* For $\theta \in [0, \pi]$, the point $(\cos \theta, \sin \theta)$ is at distance $\theta$ along the circle from $(1, 0)$, counterclockwise, so it lies on the upper half of the unit circle, $\{(x_1, x_2) : x_1^2 + x_2^2 = 1,\ x_2 \geq 0\}$, which gives $\sin \theta \geq 0$. Different values of $\theta$ in $[0, \pi]$ are different distances along that half circle, so they give different points. Two different points of the upper half have different first coordinates, because $x_2 = \sqrt{1 - x_1^2}$ is determined by $x_1$. So cos is one-to-one on $[0, \pi]$. Every $x_1 \in [-1, 1]$ is the first coordinate of the point $(x_1, \sqrt{1 - x_1^2})$ of the upper half, and the distance along the half circle from $(1, 0)$ to that point is some $\theta \in [0, \pi]$, so cos takes every value in $[-1, 1]$. Finally, move along the upper half circle from $(1, 0)$ to $(-1, 0)$ as $\theta$ increases from 0 to $\pi$. The first coordinate never takes the same value twice, so it never turns back. It starts at $\cos 0 = 1$ and ends at $\cos \pi = -1$, so it only decreases. □